Arul Rhik Mazumder

U.S. Citizen

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Research interests. What quantum algorithms really cost: lower bounds on the T gates and ancilla qubits any circuit must use, cheaper fault-tolerant algorithms such as Hamiltonian simulation, and when quantum methods genuinely beat classical ones.

Education

Imperial College London **Sep. 2026 – Sep. 2027 (expected)**
MSc, Physics with Quantum Dynamics *London, UK*

Carnegie Mellon University **Aug. 2023 – May 2026**
B.S. in Computer Science; ML and Algorithms & Complexity concentrations; University Honors *Pittsburgh, PA*

- Coursework: Quantum Computing, Complexity Theory, Learning Theory, Randomized Algorithms, Convex Optimization, Parallel Computer Architecture.

Research Experience

Institute for Quantum Computing (IQC), University of Waterloo **May 2026 – Aug. 2026**
Undergraduate Researcher (Mitacs Globalink Scholar), hosted by Michele Mosca and Vlad Gheorghiu *Waterloo, ON, Canada*

- Proved that Campbell and Howard's $6m + 1$ construction for a layer of m disjoint CCZ (Toffoli) gates is T-optimal among Hadamard-free circuits, via a lower bound computable in $O(n^3)$ time for every diagonal Clifford+T gate. Single-author, arXiv:2610.01024; posters at QRE and UnitaryCon 2026.
- The bound certifies 193 of 311 phase blocks in PyZX/TODD optimizer output as T-optimal, against 113 for stabilizer nullity, and recovers the known optimum of 17 T gates for the GF(4) multiplier.
- Completed IQC's USEQIP program: quantum information coursework and 30+ hours of experiments on quantum hardware.

Magic State Labs **Aug. 2026 – Oct. 2026**
Research Consultant, with Pradeep Niroula *Remote*

- Built soundness rules for screening candidate problems for decoded quantum interferometry (DQI) and ran an AI-agent search under them that found two new candidate applications (arXiv:2610.06753).

Caltech Institute for Quantum Information and Matter (IQIM) **May 2025 – Aug. 2025**
SURF Undergraduate Researcher, Preskill group, with Samson Wang and James Watson (Google Quantum AI) *Pasadena, CA*

- Developed early fault-tolerant algorithms for matrix functions that use commutator scaling to cut the cost of product-formula routines (arXiv:2608.13862; TQC 2026 poster).

BlueQubit **Oct. 2025 – Present**
Part-time Researcher and Hackathon Lead, AI and Core teams *Los Angeles, CA (remote)*

- Benchmarked GPU-accelerated MPS and Pauli-path simulators against AWS Braket, Quantum Rings and open-source backends; the runtime study (arXiv:2607.09882) informed product decisions and simulator cost estimators.
- Write technical blog posts and tutorials, and mentor and judge the BlueQubit track at student quantum hackathons.

Quantum Technologies Group, Carnegie Mellon University **May 2023 – Jun. 2025**
Researcher with Sridhar Tayur *Pittsburgh, PA*

- Wrote a tutorial on solving QUBO problems with QAOA and quantum annealing, with five open notebooks (*Tutorials in Operations Research*, INFORMS 2025).
- Implemented semidefinite-programming formulations for entanglement detection (CVXPY, PICOS, Gurobi).

Algorithms Group, George Mason University **May 2022 – Jun. 2025**
Researcher with Fei Li *Remote*

- Three papers on quantum and hybrid algorithms: string matching in Qiskit (AISS 2022); a hybrid D-Wave algorithm for the traveling salesman problem that beat pure classical and pure quantum baselines (IPDPS 2023); and shipment rerouting with up to 1,000 shipments on real transport networks (JSSR 2024).

Publications

Preprints

1. **A. R. Mazumder**. "Exact T-counts of CCZ Layers from an Isotropy Bound." arXiv:2610.01024, 2026.
2. **A. R. Mazumder**. "Ancilla-free Randomized Toffoli Gates and the T-count of Boolean Oracles." arXiv, submitted 2026.
3. **A. R. Mazumder**, P. Niroula. "Discovering New Problems for Decoded Quantum Interferometry." arXiv:2610.06753, 2026.
4. **A. R. Mazumder**, J. Watson, S. Wang. "Resource-Efficient Quantum Eigenvalue Transform with Commutator Scaling." arXiv:2608.13862, 2026; TQC 2026 poster.

5. **A. R. Mazumder**, H. Karapeytan, M. Z. Mullath, R. Tanin, H. Gharibyan, H. Tepanyan. "Benchmarking Zero-Setup Quantum Circuit Simulators." arXiv:2607.09882, 2026.
6. F. Li, **A. R. Mazumder**, M. Zhao. "Quantum Annealing Approaches to Shipment Rerouting Problems." arXiv:2501.05624, 2025.

Peer-reviewed

1. **A. R. Mazumder**, E. Towe. "An End-to-End Hybrid Quantum–Classical Sampling Workflow for Discrete Markov Random Fields." *IEEE QCE*, 2026.
2. **A. R. Mazumder**, S. R. Mazumder. "Depth-Efficient Quantum Topological Data Analysis for Regime-Specific Detection of Financial Stress." *IEEE QCE*, 2026.
3. **A. R. Mazumder**, S. Tayur. "Five Starter Problems: Solving Quadratic Unconstrained Binary Optimization Models on Quantum Computers." *Tutorials in Operations Research*, INFORMS, 2025, pp. 145–183.
4. M. Zhao, **A. R. Mazumder**, F. Li. "Improved Classical and Quantum Algorithms for the Shipment Rerouting Problems." *J. Student-Scientists' Research*, 2024.
5. **A. R. Mazumder**, A. Sen, U. Sen. "Benchmarking Metaheuristic-Integrated QAOA Against Quantum Annealing." *LNNS*, 2024.
6. F. Li, **A. R. Mazumder**. "An Adaptive Hybrid Quantum Algorithm for the Metric Traveling Salesman Problem." *IPDPS*, 2023.
7. M. Gao, R. Huang, **A. R. Mazumder**, F. Li. "Comparisons of Classic and Quantum String Matching Algorithms." *AISS*, 2022.

Talks and Posters

- **QRE 2026** and **UnitaryCon 2026**, Toronto — posters, "Certified T-count floors for Toffoli networks and reversible circuits"; **IEEE QCE 2026**, Toronto — two first-author papers.
- **TQC 2026** — poster, "Resource-Efficient Quantum Eigenvalue Transform with Commutator Scaling" (PDF).
- **Caltech IQIM SURF Seminar**, Aug. 2025, and **Quantum Leap Career Nexus**, UMD, Nov. 2025 — talks, "Early Fault-Tolerant Quantum Algorithms for Matrix Functions via Trotter Extrapolation" (slides).
- **INFORMS Annual Meeting 2025** — talk on solving QUBO models on quantum computers.

Awards and Honors

- **Mitacs Globalink Research Award** and **IQC Undergraduate Research Award** (2026); **IEEE QCE Student Travel Grant** (2026).
- **Duke STAQ NSF-funded Invited Scholar** (2025): PhD-level week on quantum simulation, error correction and architecture (Peter Love, Kenneth Brown, Yu Tong).
- **MIT iQuHACK** (2026): 2nd of 90+ teams, NVIDIA track; variational LABS with Pauli correlation encodings on GPUs.
- **YQuantum** (2025, Yale): Yale Quantum Institute Grand Prize, 1st in the Alice & Bob track, 2nd overall; Wigner-function state reconstruction with convex optimization and CNN denoising.
- **Citadel/Correlation One Datathon** (2024): 2nd of 150 invited from 3,000+ applicants; pairs trading, Sharpe 1.58.
- **HackCMU** (2023): GSA Infrastructure Prize for Eco-Bin, an AI waste sorter piloted with CMU Dining.
- **Regeneron Science Talent Search Scholar** (2023), top 300 of 2,000+ nationally. **Olympiads:** USA(J)MO qualifier, USAPhO Bronze, USACO Gold, USNCO National Finalist.

Leadership

CMU Quantum **Sep. 2024 – May 2026**
Founder and President *Pittsburgh, PA*

- Founded CMU's first student quantum organization (200+ members, 10+ student leaders); secured sponsorship from General Motors and Boeing; ran workshops with IBM, IonQ and Quantinuum; wrote teaching materials, led hackathon tracks and organized sponsored research trips.

Technical Skills

Languages and proof tools: Python, Julia, C++, Lean 4
Quantum and optimization: Qiskit, PennyLane, CUDA-Q, D-Wave Ocean; CVXPY, PICOS, Gurobi; CUDA, MPI